

HARMAN PREET SINGH

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ACADEMIC EMPLOYMENT

University of Tübingen	<i>Nov. 2025 - Present</i>
Post-doctoral researcher in the groups of Marius Lemm and Stefan Teufel	Tübingen - Germany
Scuola Internazionale Superiore di Studi Avanzati (SISSA)	<i>Oct. 2021 - Oct. 2025</i>
Ph.D. student in the group of Marcello Porta	Trieste - Italy

EDUCATION

Scuola Internazionale Superiore di Studi Avanzati (SISSA)	<i>Oct. 2021 - Oct. 2025</i>
Ph.D. in Geometry and Mathematical Physics	Trieste - Italy
Advisor: Marcello Porta (Mathematics Area)	
Thesis: <i>Large-scale response theory for gapless Fermi systems in low dimensions</i>	
Final Mark: Ph.D. awarded cum laude	
Università degli Studi di Padova	<i>Oct. 2019 - Sep. 2021</i>
M.Sc. in Physics	Padova - Italy
Advisors: Antonio Ponno & Lorenzo Zanelli (Mathematics Department)	
Thesis: <i>A rigorous derivation of the Bogoliubov-Gross-Pitaevskii equation for superfluids</i>	
Final Mark: 110/110 cum laude	
Università degli Studi di Padova	<i>Oct. 2016 - Sep. 2019</i>
B.Sc. in Physics	Padova - Italy
Advisor: Pieralberto Marchetti (Physics Department)	
Thesis: <i>Quantum Mechanics in phase space: non-classical properties</i>	

PUBLICATIONS AND PREPRINTS

2. M. Porta, G. Scola, and H. P. Singh, *Large scale dynamical response of interacting 1d Fermi systems*, (2025) arXiv: [2509.08665 \[math-ph\]](https://arxiv.org/abs/2509.08665), to appear on *Comm. Math. Phys.*
1. M. Porta, and H. P. Singh, *Large Scale Response of Gapless 1d and Quasi-1d Systems*, Annales Henri Poincaré (2025) <https://doi.org/10.1007/s00023-025-01600-z>

TEACHING EXPERIENCE

Advanced topics in Mathematical Quantum Theory	<i>SS 2026</i>
Teaching assistant for prof. M. Lemm, MSc. in Mathematical Physics program	U. Tübingen
Mathematical Quantum Theory	<i>WS 2025</i>
Teaching assistant for prof. S. Teufel, MSc. in Mathematical Physics program	U. Tübingen

CONFERENCE TALKS AND POSTER SESSIONS

- 13/2/25 **Mathematical Challenges in Quantum Mechanics**, GSSI, L'Aquila, Italy.
Contributed talk: *Large scale edge response for 2d topological insulators*
- 16/12/24 **127th Statistical Mechanics Conference**, Rutgers University, Piscataway, New Jersey, USA.
Contributed talk: *Validity of edge linear response for 2d topological insulators*

- 29/8/24 **Rigorous Renormalization Group Analysis of Collective Phenomena in Fermionic Quantum Systems**, Como, Italy.
Contributed poster.
- 13/8/24 **Quantissima in the Serenissima V**, Venice, Italy.
Contributed talk: *Validity of edge linear response for 2d topological insulators*

SEMINAR TALKS

- 15/12/25 **Junior SFB Coffee Talk**, online seminar
Validity of Kubo formula for 1d and quasi-1d lattice fermionic systems
- 13/11/25 **Mathematical Physics Oberseminar**, University of Tübingen, Tübingen, Germany
Large-scale response theory for gapless lattice Fermi systems in low dimensions
- 30/1/25 **Mathematical Physics Oberseminar**, University of Tübingen, Tübingen, Germany
Large scale response of gapless 1d and quasi-1d fermionic systems
- 13/3/23 **Junior Geometry and Mathematical Physics Seminar**, SISSA, Trieste, Italy
A geometric picture of topological phases of matter
- 1/6/22 **Renormalization Group seminar**, SISSA, Trieste, Italy
Determination of the RG fixed point via tree expansion
- 31/5/22 **Seminar-exam**: Riemann surfaces and integrable systems, SISSA, Trieste, Italy
The canonical ensemble of the periodic Toda lattice

SCIENTIFIC VISITS

University of Zürich (UZH) Hosted by: B. Schlein and M. Porta	<i>May 2024</i> 1 week
University of Zürich (UZH) Hosted by: B. Schlein and M. Porta	<i>March 2024</i> 1 week
University of Zürich (UZH) Hosted by: B. Schlein and M. Porta	<i>Nov.-Dec. 2023</i> 1 week

SERVICE ACTIVITIES

Member of SFB Junior Committee within Collaborative Research Center TRR 352 <i>Mathematics of Many-Body Quantum Systems and Their Collective Phenomena</i> , joint venture of LMU and TU in München, and U. Tübingen	<i>Nov 2025 - Present</i>
Co-organiser of the monthly SFB Junior Seminar	<i>Nov 2025 - Present</i>

LANGUAGES

Italian	Native
Punjabi	Native
English	Fluent (level C1 on the CEFR for Languages)
Hindi	Fluent
French	Basic knowledge

RESEARCH INTERESTS

My research interests focus on the mathematical analysis of transport in quantum many-body systems, performed by employing techniques drawn from Quantum Field Theory and equilibrium Statistical Mechanics. Further, I am keen on the emergence of topological and geometrical invariants parameterising robust phases of matter.

During my PhD, I worked on the proof of validity of linear response, encoded in Kubo formula, for *gapless* lattice fermionic systems in low dimensions, via rigorous Renormalisation Group techniques.

Currently, I am investigating the extension of these results to higher dimensional models. Furthermore, I am studying the information propagation properties of fermionic systems in the continuum setting.